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RESEARCH ARTICLE

Intelligent Gesture-Enhanced Blockchain Voting: A New Era of Secure and Accessible E-Voting

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ABSTRACT Electronic voting systems have often associated with issues of security, transparency, and limited accessibility, particularly for people with disability, elderly, and visually impaired voters. These systems, impeded by complex authentication procedures and non-intuitive user interfaces, have also struggled to ensure voter trust and widespread participation. To address these issues, this paper presents a new electronic voting system that combines the gesture recognition with blockchain technology. The proposed system has two phases, the first one is a user identification phase using a pre-trained dataset based on both cvzone and Dlib tools for face and finger identifications. Gestures from the face and hands, such as eyebrow and nose movements are used as authentication mechanisms as well as a mean of input into the system. The second is a voting process phase which uses the blockchain technology. This new approach strengthens the security and transparency of voting through a tamper-proof ledger provided by the blockchain technology, as well as enhances considerably the inclusiveness and also user friendliness. Real time simulations show that the user interaction times for different age groups ranges between (24.3s to 29.7s) and the accuracy rate for gesture types varies between (88.7s to 92.5s). Moreover, the simulation results indicated that real time voting using our proposed system generated 100% precision and 99% recall. Our system showed high accuracy in gesture recognition and profound security when using Equal Error Rate quality measure. Different to other approaches, the interface of our proposed system, designed for a wide audience of users, greatly simplifies the voting process.

INDEX TERMS Blockchain, electronic voting, gesture recognition, accessibility, security, user interface.

I. INTRODUCTION

In contemporary times, electronic voting systems (EVSs) have emerged as major tools to ensure the efficiency and modernization of the electoral process. On the other hand, these systems have several challenges regarding their security and integrity [1], [5]. In many countries around the world, transparent elections are increasingly being challenged and compromised [6], [9]. The current design of the digital voting

systems presents many barriers that prevent inclusivity and accessibility for a diverse electorate where, user authentication is a key component of the EVS [10], [11]. Many voters could encounter physically challenged including elderly voters, as well as visually impaired, whose interaction with technology does not accord to the conventional methods may find the complex authentication procedures a barrier. In addition, the complexity of these systems can discourage the participation because they do not provide enough support to the people who need assistive technologies. Moreover, the user interfaces of the traditional e-voting systems often lack

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intuition, which only drives the voters away by creating an impression that it is difficult to use for the electorate. The latest studies show that it is possible to enhance the security through the adoption of electronic voting techniques [12], [16]. Electronic voting systems encounter several critical challenges that need to be addressed to improve their inclusivity, security, efficiency, and real-time performance. Firstly, the majority of current electronic voting systems depend on a single type of biometric data, which limits their ability to accommodate the diverse attributes of voters. Disabled and elderly voters often encounter substantial barriers when using electronic voting systems. Many of these systems do not adequately address the physical limitations of these groups, making participation in the voting process challenging. For instance, traditional biometric methods like fingerprint scanning or facial recognition can be difficult for individuals with physical disabilities or visual impairments. Additionally, the complexity of these systems can deter elderly voters who may not be familiar with modern technology, thereby reducing overall voter participation. This limitation raises concerns about the system's inclusivity for all users, particularly in national election contexts. Furthermore, integrating biometric methods with blockchain technology to enhance security, privacy, scalability, and inclusivity presents significant challenges. Finally, blockchain systems struggle with real-time processing speeds during peak periods, such as national elections, leading to potential delays. [17], [20].

Consequently, our primary motivation is to improve the inclusivity, security, efficiency, and real-time performance of electronic voting systems by incorporating gesture-based biometric inputs with blockchain technology. This method addresses major challenges faced by disabled and elderly voters, who often encounter barriers with traditional biometric systems, and aims to enhance voter participation and system accessibility.

Thus, the main contributions of this article are presented as follows:

- 1) This paper presents a novel voting method that combines gesture recognition with blockchain technology to tackle the issue of limited biometric authentication in electronic voting systems. By accommodating various biometric types, this approach improves system accessibility for a broad spectrum of voters. Gesture authentication eases access, and blockchain secures vote recording, marking a significant leap towards a more secure, inclusive, and user-friendly electronic voting system in the digital age.
- 2) The proposed system is superior to the common voting methods in many ways most notably by significantly enhancing the access of disabled persons, the elderly and also those with visual impairments. It offers a virtual voting system that enables distant participation for the physically challenged voters like the senior citizens and also disabled individuals. By implementing non-touch gestures, the system reduces physical exertion, making it easier for elderly users and those with physical

disabilities to authenticate their identity and cast their votes.

- 3) This paper introduces a user-friendly interface for electronic voting systems that overcomes the challenges of gesture-based biometric inputs, making voting simpler and more accessible. The design prioritizes intuitiveness and ease of use for everyone, including those with limited tech skills or physical disabilities affecting gesture use, and it specifically caters to individuals with cognitive limitations. This approach greatly increases the inclusivity and convenience of electronic voting, encouraging wider participation from various voter groups.
- 4) The proposed system transforms the voter authentication process through its use of gesture-based verification, eliminating the need for complex passwords or PINs. It shows adaptability by allowing multiple gesture inputs, which accommodate various user preferences and physical limitations. Additionally, non-touch gestures help reduce physical exertion, aiding elderly users and providing inclusive authentication methods, such as eyebrow or nose movements, for those who struggle with conventional biometrics.
- 5) The proposed system addresses the challenge of real-time processing speeds in blockchain-based systems, particularly during peak times such as national elections. These enhancements ensure timely vote recording and processing without delays.

These innovations substantially enhance the inclusivity, security, efficiency, and real-time performance of electronic voting systems, thereby enabling wider and more secure voter participation in the digital era.

The reminder of this paper is structured as follows. Section II discusses related research including blockchain based voting systems, and intelligent authentication methods. In Section III, the methodology and methods used in this research will be explored, while Section IV discusses more details about Gesture Recognition and Blockchain-based Voting System. Section V, the results will be shown and comparison with the state-of-the-art techniques will be provided. Finally, Section VI, provides conclusion and future works.

II. RELATED WORK

In this section, various studies that integrate blockchain technology into electronic voting systems to enhance security and transparency will be explored.

A. BLOCKCHAIN-BASED VOTING SYSTEMS

Blockchain technology has been considered a viable solution for safe and transparent electronic voting systems. By exploring the benefits of blockchain technology such as decentralized management, immutability, and transparency, electronic voting systems can reduce manipulation and fraud, enhance voter anonymity, and enhance confidence in the election process.

Pereira et al. [21] explored the potential of blockchain technology to address prevalent challenges in security and

TABLE 1. Evaluation of recent blockchain voting methods: advantages and constraints.

Study	Method	Advantage	Limitation
[21]	Propose a mixed-methods strategy that includes genetic algorithms, blockchain technology, and user voting.	Their system aims to optimize scheduling and evaluate blockchain integrity.	Difficulty in authentication processes, especially for individuals with physical disabilities.
[22]	Present a model that uses blockchain technology, specifically the Ethereum network, Ganache blockchain, and the Solidity programming language.	The paper highlights, to process the flaws and limitations of the traditional paper-based voting system.	The main disadvantage, is complex interfaces, hence reducing voter turnout.
[23]	Using Blockchain, with flexible consensus algorithms, and smart contracts.	Provide transparency and reliability in the voting system.	difficulty of accessibility particularly for individuals who may not have access to technology or face barriers to using it.
[24]	Present a technique that using Smart contracts.	The use of smart contracts in a permissioned blockchain ensures the authenticity and verifiability of votes.	It does not include any alternative method of voting.
[25]	The proposed model uses blockchain technology, specifically Ethereum 2.0, with Solidity as a programming language for generating smart contracts.	addresses the limitations of traditional paper balloting and digital voting methods.	It does not implement the OTP (One Time Password) option in the registration process.

transparency across various sectors, with a specific focus on electoral systems and academic scheduling.

This goal was achieved by utilizing a mixed-methods strategy that includes genetic algorithms, blockchain technology, and user voting to choose optimum schedules and evaluate blockchain integrity. The research incorporated sophisticated cryptographic techniques, such as digital signatures, homomorphic encryption, and zero-knowledge proofs. However, complex authentication processes in voting pose challenges for individuals with physical disabilities, inclusive of visual, auditory, or mobility impairments. For this reason, the interfaces of digital devices and voting applications might not be designed adequately to meet their specific needs.

Vinsura et al. [22] highlighted the significance of voting as a fundamental part of democratic decision-making. It acknowledges the traditional paper-based voting system's limitations, such as cost. The authors propose blockchain technology as a solution to these issues due to its inherent security and decentralization. Their system presents a safe voting system created using blockchain technology, specifically the Ethereum network, Ganache blockchain, and the Solidity programming language. The gap identified by their paper shows how complicated the interfaces of current e-voting systems may repulse less tech-savvy individuals, hence reducing voter turnout.

The work presented in [23] focuses on blockchain technology-based solutions to the voting systems to overcome the problems of widespread mistrust in traditional voting methods and the lack of transparency provided by digital alternatives. The core problem is in the system's susceptibility to manipulation, which undermines the citizen's fundamental rights. To address these issues, their system utilizes a scalable blockchain, flexible consensus algorithms, and smart contracts.

The paper did not address the potential concerns related to the accessibility and inclusivity of a digital voting system, particularly for individuals who may not have access to technology or face barriers to using it.

The research shown in [24] explores the capacity of the usage of the blockchain era to broaden digital voting structures as a service. Traditional digital voting systems have confronted practical demanding situations, and blockchain's attributes of immutability, encryption, and safety present a promising solution. Their study discusses the targets of electronic voting systems, highlights technological and criminal obstacles of blockchain as a service, and introduces a novel blockchain-based vote-casting gadget that addresses current shortcomings. By employing smart contracts, their machine complements security, reduces costs, and preserves voter anonymity. However, the gap identified in their paper is the lack of flexibility in voting interfaces to suit different abilities, creating unfairness in voting, and the absence of alternative methods in their system.

The work shown in [25] discusses challenges in traditional and digital voting systems, noting vulnerabilities like errors and potential rigging.

It proposes a blockchain-based solution for its robust verification capabilities, aiming to ensure anonymity, privacy, verifiability, and fairness. The system anonymizes voter data using blockchain hashes, encrypts votes for fairness, and enables post-election vote verification. Implemented on Ethereum 2.0 with Solidity for smart contract development, it enhances voter verification, accuracy, transparency, and security against fraud. Their paper highlights the system's lack of a one-time password (OTP) for registration, which poses challenges for elderly voters, and non-interactive interfaces, leading to lower voter turnout.

Blockchain technology has substantially enhanced data integrity and security, enabling creative solutions for a wide range of applications. Notable examples include LedgerDB and VeDB, which are blockchain-based databases with great performance and tamper protection. LedgerDB, developed by Alibaba Cloud, incorporates cryptographic approaches to offer tamper-evidence and non-repudiation, making any attempt to alter past data apparent. This makes it ideal for applications that require efficient and safe data handling [26].

TABLE 2. Biometric and multi-factor authentication in e-voting: A comparative analysis.

Study	Method	Advantage	Limitation
[28]	The methodology combines secure authentication algorithms on smart cards and fingerprint biometrics to achieve a reliable MFA technique.	The goal is to develop novel countermeasures to meet voter' authentication, vote confidentiality.	The system is not suitable for voters with physical limitations that make it difficult for them to interact with the electronic voting systems.
[29]	Proposes an online voting system that incorporates features such as election administration, survey administration, and facial authentication, using Laravel PHP framework and Python deep learning for face authentication.	Their system highlights, ensuring transparency and minimizing the possibility of manipulation.	Using multi-factor authentication, which might lead to accessibility problems and also restrict system inclusivity.
[30]	Proposes an electronic voting authentication scheme that utilizes multiple keys (global, pairwise, and individual).	Providing a comprehensive and secure key management mechanism that addresses the diverse security needs of the voting network	Increased Complexity, and also Biometric verification may not be available to everyone.
[31]	The paper presents model by using fingerprint and visual semagram techniques.	Offers a secured and reliable method for conducting elections, and also reduce concerns associated with traditional paper-based voting systems	The main disadvantage is difficulty in authentication.
[32]	The paper Proposed a technique, utilizes biometrics-based facial and fingerprint detection using image processing and Convolutional Neural Networks.	It reduces the time-consuming manual labor involved in traditional voting systems.	Using multi-factor authentication, which can be challenging for some users.

Similarly, VeDB is a dependable relational database that combines software and hardware security components to provide tamper-proof functionality while preserving high performance and verifiability. VeDB employs blockchain-like structures to keep data immutable and verifiable, making it ideal for applications that require high data integrity and security [27]. Despite their broad capabilities, neither LedgerDB nor VeDB have been explicitly used in voting systems.

B. INTELLIGENT AUTHENTICATION METHODS

Security has become an important consideration in managing our accounts in all fields. In most computer security scenarios, user authentication is an essential component.

The researchers in [28] proposed a multi-factor authentication (MFA) technique, making use of fingerprint biometrics and a cryptographically secured smart card. MFA combines a more suitable Feistel block cypher set of rules and a first-moment feature extraction technique to reinforce the safety of the e-balloting machine's authentication method. The results shown in their study constitute a critical step in the improvement of a practical and steady e-balloting device for vote confidentiality. Furthermore, the challenges outlined in their paper include the potential unsuitability of fingerprint biometrics and smart cards for users, particularly for those with physical disabilities. Piam et al. [29] developed an internet voting device composed of three main parts: election management, survey management, and face authentication. The system is implemented in Laravel, a PHP framework for web applications, with the support of deep learning for facial recognition. The author further says that the developed online voting machine provides a stable and user-friendly participative environment with software for election, survey, and announcements, and, at the same time,

has facial authentication for voter identification. According to the authors, such a developed system will be reliable and provide a democratic process of effective interaction. Mainly, the issues that have been emphasized in the paper are related to a heavy dependence on facial authentication, which may fail to recognize the user with major changes from surgery or medical treatment, and the multi-factor authentication may create the problem of accessibility and even restrict the system's inclusiveness. In [30], the authors stated that they have introduced a new e-voting system using a biometric verification process for the voter authentication. When authenticated, authorized voters' votes are cast securely with their own data transmitted to the server. The e-voting system developed by them uses the EVAS, which overcomes the weaknesses of single-key encryption schemes and also improves security. EVAS provides multiple keys per polling station, including Global Keys (GK), Pairwise Keys (PK) and also Individual Keys (IK) for different messages and communication requirements. The limitations in their paper, represented by biometric verification which can be a challenge for individuals with skin conditions or injuries, and the requirement of multiple keys may complicate the process for some users, particularly for those with cognitive impairments or limited digital literacy. Adewale et al. [31] address the need for a more secure and reliable voting system that combat fraudulent activities that compromise the integrity of elections. The study makes improvement of an electronic vote system gadget that utilizes fingerprint and visible semagram strategies to enhance security.

Adewale et al. proposed e-vote model includes six parts designed for effectiveness inside the electoral system. The gadget was tested in an authentic deanship election involving five faculties at a tertiary institution. An eligible team of workers participants were enrolled and provided with a

voter identity range (VIN) card, with voter authentication counting on fingerprints and VIN. To ensure data safety, sensitive election outcomes have been encrypted and hidden within an image, their system was developed using the visual semagram technique. The limitations of this work involve relying on fingerprint and visual semagram methods which can be cumbersome or inaccessible for some users; also, the need for a VIN card adds another step, leading to lower participation among users who find it too complex. In [32], the authors proposed a web-based system utilizing biometrics-based facial and fingerprint detection using image processing and Convolutional Neural Networks.

Voters sign in on an online platform by providing their name, address, fingerprint, and facial photo. The Election Commission collects and shops this information securely. On Election Day, voters' login to the system using their faces and fingerprints, similar to how mobile phones are unlocked. Their system also employs a unique feature, the space between the eyes and eyebrows, which stays steady with age and, utilizes a tenprint image method to detect the correct name of the voter, decreasing the possibility of fraudulent vote casting. The challenges in their work involve the dependence on facial biometrics and fingerprint recognition which can be problematic for voters with facial abnormalities or skin conditions.

Unfortunately, the presented challenges in existing e-voting systems point to a lack of inclusivity, accessibility, and user-friendliness, potentially disenfranchising groups with disabilities, elderly, and less tech-savvy individuals. These drawbacks range from complex authentication to unreliable biometrics, cumbersome interfaces, and insufficient security measures, ultimately leading to lower voter turnout and trust in the electoral process. Hence, we proposed a novel voting system that solves the indicated disadvantages using innovative measures such as simplified authentication, adaptive interfaces, increased transparency, and also user-centered design.

III. PROPOSED METHODOLOGY

In this paper, we are presenting a new voting system that integrates the inputs provided through gestures detected by vision via a camera with blockchain technology. Blockchain is a revolutionary technology that provides a decentralized way for safely and transparently storing data. It consists of a chain of encrypted blocks that are connected together, making data manipulation practically difficult while increasing transparency and credibility.

Users vote by interacting with a virtual form. Voting inputs are provided through distinct gestures: The Dlib tool counts raising the right or left eyebrow or nose, signaling with the finger that is picked up through vision using a camera without physically touching the screen. These gestures provide creative and simple approach to the voting procedure. Whenever a vote is successfully submitted, the data gets secured on a blockchain platform, whereby each vote becomes permanently, transparent and immutable. The combination of

gesture detection through a camera and the blockchain's unparalleled recording leads to an innovative way toward the transparent, user-centric voting system. As shown in Figure 1.

Initially, the activation process of the proposed system begins with configuring the camera to enable real-time video capture, along with displaying the virtual keyboard on the screen using the following steps.

Step1: Users are granted access to the system through the use, of one of the various biometric inputs, such as left or right eyebrow movement, the nose interaction, or fingerprint recognition to enter the system.

Step2: The system undergoes a verification procedure against an existing database to authenticate the entered information. In cases where initial verification fails due to discrepancies in the provided information, the system prompts the user to re-enter their data using step 1.

Step 3: Upon successful verification in step 2, confirming the user's legitimacy as a voter, the system proceeds to the voting interface where available options are presented.

Step 4: A new transaction is generated, containing the voter's identification and selection, which is then incorporated into a blockchain through the creation of a new block. This block encapsulates the transaction, ensuring the integrity and immutability of the voting data, subsequently stored in the database.

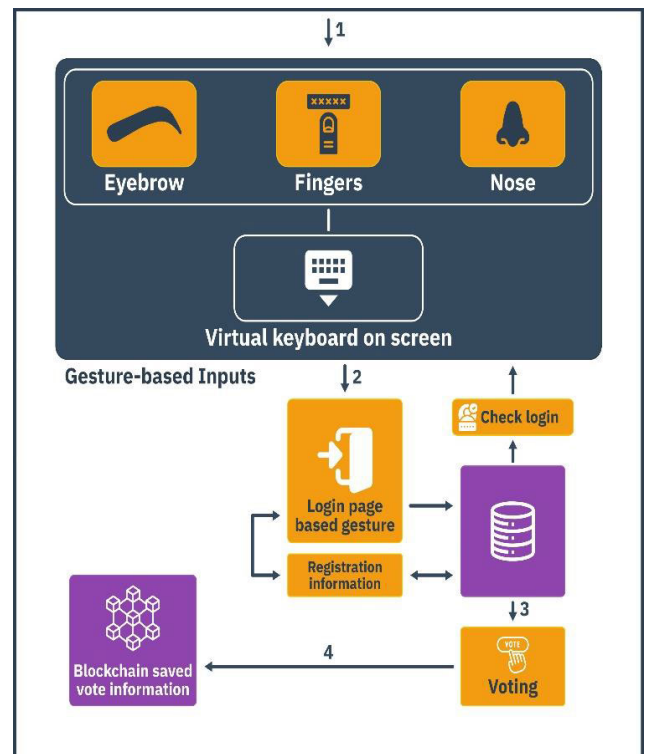


FIGURE 1. Proposed voting system block diagram.

IV. GESTURE RECOGNITION

The gestures retrieved from the face incorporate right and left eyebrow, nose, and finger gesture. These actions are used as

inputs to the system. It is essential to understand the steps of Gesture Recognition for both cases as follows.

The Gesture Recognition System leverages pre-trained models from dlib and cvzone libraries to accurately track recognition of gestures made by the face along with hand gestures. Dlib and, in particular, can leverage the latest machine-learning technologies to find the human face on a video feed and map 68 different facial landmarks. It enables the localization of several facial gestures with respect to movement of eyebrows or nose by tracking the relative positioning of these landmarks; at the same time, cvzone develops its strong pre-trained model for hand gesture recognition. The model locates and finds the existence of hands in the video feed, recognizing pinching by measuring the distance between the tips of the thumb and forefinger. With these technologies integrated into the system, it “translates subtle facial and hand movements into input for interactive applications, such as virtual keyboards, hence offering a new, intuitive way of using natural gestures. This leads to synergy with pre-trained models, ensuring the responsiveness of gesturing recognition precision, so that many of the usability factors and high levels of accessibility of the system are achieved [33].

A. FACE GESTURE RECOGNITION-USING Dlib

Face Gesture Recognition by Dlib is a technique used to detect and analyze facial movements or expressions using computer algorithms. Dlib library that is known for its capabilities in machine learning and computer vision uses facial data to identify various types of gestures themselves. This technology is particularly useful in the areas of user authentication, virtual gaming, and also accessibility tools that involve real-time tracking of facial gestures for immediate human computer interaction [34]. Figure 2 (part 1), shows the process of gesture recognition using Dlib tool and involves the following steps:

1. Face Image Acquisition: This step involves taking the user's face image using a camera.
2. Face Image Processing: The acquired image undergoes processing.
3. Eyebrow Recognition and Extraction: The eyebrows are identified and their characteristics are determined.
4. Nose Recognition and Extraction: The nose, especially its tip is identified and its features are exacted.
5. Feature Extraction: Extraction of the unique properties of the recognized features (eyebrows and nose) for additional analysis as well as gesture recognition.

While Figure 3 shows the steps of Face Gesture Recognition using Dlib utilized in our proposed voting.

B. HAND GESTURE RECOGNITION - USING CV ZONE

Hand Gesture Recognition through the cvzone utilizes computer vision in recognizing and analyzing hand gestures. Using the cvzone library based on OpenCV, this method allows for real-time tracking of hand gestures. It is a popular tool for authoring interactive applications such as virtual

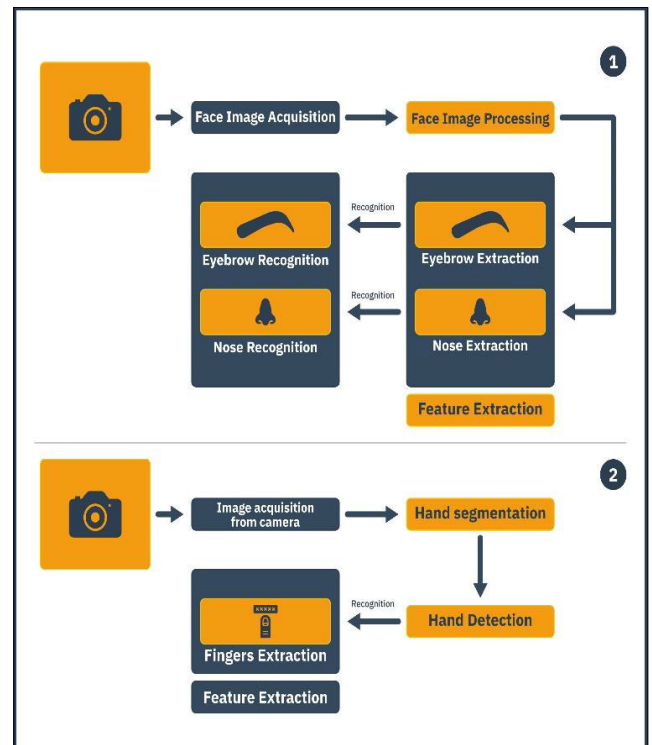


FIGURE 2. The process of gesture recognition. (a) Part 1: Implementation of Dlib for detecting facial landmarks and recognizing initial gestures. (b) Part 2: Utilization of Cvzone for advanced gesture analysis and final classification.

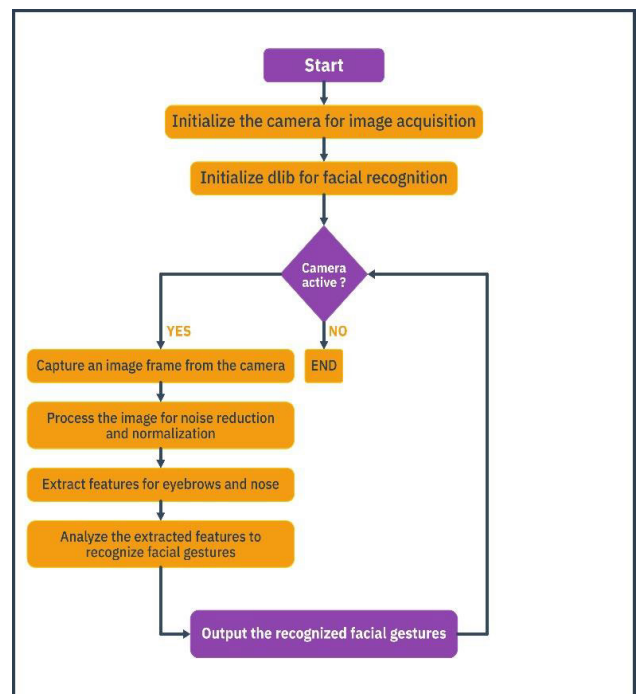


FIGURE 3. Shows the steps of face gesture recognition using Dlib.

reality systems, sign language interpreters and user interface controls since it allows the users to easily interact with the

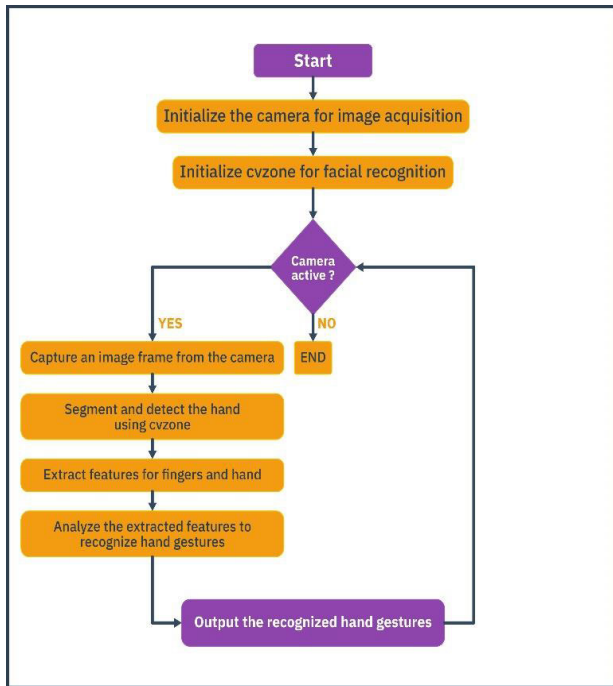


FIGURE 4. Shows the steps of hand gesture recognition using cv zone.

digital platforms [35]. Figure 2 (part 2), shows the process of gesture recognition using CV zone as follows:

1. Image Acquisition from Camera: this step takes the user's hand image.
2. Hand Segmentation: the acquired image undergoes segmentation to isolate the hand from the rest of the image.
3. Fingers Extraction: Features of the fingers are extracted.
4. Hand Detection: The entire hand is detected and recognized.
5. Feature Extraction: The unique characteristics of the recognized features (fingers and the hand) are extracted for further analysis and gesture identification.

While Figure 4 demonstrates the steps of Face Gesture Recognition using cv zone utilized in our proposed voting.

C. GESTURE-BASED INPUT

Gesture-based input systems offer a seamless method for users to interface with devices through physical gestures. These can be hand motions, finger movements, or facial expressions. Instead of relying on conventional devices like keyboards, mice, or touchscreens, users can employ intuitive body movements to provide input. Within the context of the shared code, facial landmarks (both eyebrows and the nose) and hand gestures are the primary modes of interaction.

Real-time eyebrow tracking enables a unique and interactive mode of input. When a user intentionally moves either their left or right eyebrow to hover over a virtual button, the system deciphers this action as an intentional button press. For illustrative purposes, if a user navigates their left or right eyebrow over the 'Q' button, the system seamlessly translates this gesture as an equivalent to pressing the 'Q' key on a

traditional keyboard. Furthermore, the choice between using the left or right eyebrow is entirely dependent on the user's preference. They have the flexibility to select either eyebrow as their primary input mode.

The position of the nose, particularly its tip, can function as an innovative pointing or selection tool in this gesture-based system. Much like how a mouse cursor is employed to navigate and choose items on a traditional digital screen, the movement of the nose offers a similar interactive experience with virtual buttons in this setup. An intentional gesture, such as a nod, could be interpreted as a confirmation action, analogous to a "click". This functionality aligns with the flexibility provided for the left and right eyebrows. Depending on individual comfort and preference, users might find the nose, one of the eyebrows, or a combination of these to be the most intuitive and effective means of interaction. By incorporating such diverse input options, the system underscores its commitment to a tailored and user-friendly experience.

The code identifies a pinching motion by determining the gap between two fingertip landmarks. If this distance falls below a defined limit (e.g., 40 units), it is construed as a "click" or selection gesture, allowing users to interact with virtual buttons.

D. INNOVATIVE VOTING: COMBINING BLOCKCHAIN AND GESTURES

Upon successful system entry, the voting interface becomes accessible. Same gesture recognition mechanisms that aided in the authentication process are repurposed for the act of casting votes. In particular, voters can employ gestures like raising the right or left eyebrow, moving the nose and also use any finger to make their choice.

These user-friendly and intuitive approaches give power to the voter, making the vote casting seamless and user-friendly. What sets our suggested voting system apart from the traditional methods is its technological basis. With the use of blockchain technology, each vote becomes a part of an immutable and also decentralized ledger.

This guarantees unparalleled security and transparency, which eliminates any possibility of electoral fraud. Every cast vote serves as a separate block within the chain, and any attempt to modify this block is immediately observed because of the nature of the blockchain. Furthermore, the blockchain is a decentralized system where no authority controls the entire voting record, thereby preventing any form of centralized manipulations. This gives a voting system that is user-friendly, using gesture inputs, secure and reliable by default.

A cryptographic hash represents each block's distinct identity. The hash is formed based on the content of the block through the SHA-256 algorithm. The slightest alteration in the block's content, even a single character, would lead to a completely different hash. This property ensures the immutability of blocks: If someone tries to change a vote in the block, it would become obvious because of the hash mismatch.

Every blockchain starts with a ‘genesis block’, which serves as the anchor for the entire chain. This special block is hardcoded and does not reference any previous block. All subsequent blocks refer back to the genesis block through a chain of hashes.

The ‘nonce’ in a block is a variable value that miners or validators change to achieve a hash for the new block that satisfies certain criteria. This iterative process, requiring significant computational work, ensures that adding a new block is not trivial, and frustrates spam and malicious additions. The specific criterion in this case is that the computed hash must start with a certain number of zeros, determined by the system’s “difficulty”.

As votes come in, they are not immediately added to the blockchain. Instead, they go to a ‘pool’ of unconfirmed transactions. Periodically, these are grouped and processed in a batch to form a new block through the mining process, ensuring system efficiency.

In a decentralized blockchain system, there’s no central authority. All participants (nodes) in the network possess their copy of the blockchain. When nodes interact, they can share blockchain data, ensuring network-wide consistency. New nodes can register with existing ones to receive the latest copy of the chain and even inform other nodes of their presence.

With multiple nodes and potential discrepancies in their blockchain versions, a consensus mechanism is paramount. The implemented approach is straightforward: the longest valid chain in the network is considered the authoritative one. If a node realizes its version of the chain is shorter than a peer’s, it will adopt the longer version after validating its integrity. This mechanism leans on the premise that producing a longer chain requires considerable computational power, so a longer chain likely has more collective consensus.

Once a node successfully mines a new block (solves the Pow), it shares this block with its peers. Each peer can quickly verify the block’s validity by Pow and then add it to their own chain. This rapid propagation ensures the network quickly reaches a consistent state.

For users, the intricacies of blockchain are abstracted away through a voting application. This application communicates with the underlying blockchain, allowing users to cast votes, view voting results, and even see unconfirmed votes.

Each vote requires identifying data, ensuring a unique vote, and preventing double-voting. If anyone tries to alter a vote post hoc, it would require changing all subsequent block hashes, and disrupting the consensus with other nodes, making tampering evident and rendering the attempt futile.

The algorithm outlined describes a comprehensive blockchain process specifically tailored for a voting system. Here’s a step-by-step explanation of how each component works within the context:

The process begins by checking if the blockchain is empty; if so, a genesis block is created as the first block in the blockchain. Upon receiving a vote that includes the voter’s ID and choice, the system checks the entire blockchain to ensure that the voter has not already voted. If a duplicate vote

Algorithm 1 Shows the various blockchain steps.

Input:

- Vote containing voter_id and selected choice.
- A list of Unconfirmed Transactions.
- Current Blockchain containing a sequence of blocks.
- Peer nodes in the network, Peers.

Output: A blockchain with the new vote incorporated, or an error/rejection if the vote is not valid.

Begin

Step1: Initialization:

Step 1.1: Check if the Blockchain is empty.

Step 1.2: If yes, create the Genesis Block.

Step 2: Vote Casting:

Step 2.1: Receive a Vote with voter_id and selection choice.

Step 2.2: Verify if voter_id has already cast a vote by inspecting the entire Blockchain. If yes, reject and exit.

Step 2.3: Add the Vote to Unconfirmed Transactions.

Step 3: Block Mining (Execute when a certain threshold of unconfirmed transactions is reached or after a specific time interval):

Step 3.1: Select transactions from Unconfirmed Transactions to form the new block’s data.

Step 3.2: Reference the previous block’s hash.

Step 3.3: Start the Proof of Work (Pow) process:

Step 3.3.1: Initialize a nonce to zero.

Step 3.3.2: Compute the hash of the block with the current nonce.

Step 3.3.3: Verify if the computed hash meets the criterion (e.g., starts with ‘00’).

Step 3.3.4: If not, increment the nonce and return to step 3.3.2.

Step 3.3.5: Else, proceed to the next step.

Step 3.4: Finalize the block by setting its nonce and hash.

Step 3.5: Append the new block to the Blockchain.

Step 3.6: Clear the Unconfirmed Transactions that were added to the block.

Step 4: Consensus Mechanism:

Step 4.1: For each node in Peers:

Step 4.2: Retrieve their version of the Blockchain.

Step 4.3: If a node’s blockchain is longer and valid:

Step 4.4: Replace the current Blockchain with the longer one.

Step 5: Block Propagation:

Step 5.1: Announce the newly mined block to all nodes in Peers.

Step 5.2: Each peer node will:

Step 5.3: Validate the new block.

Step 5.4: If valid, add it to their blockchain.

Step 6: Output:

Step 6.1: Return the updated Blockchain.

End

is found, it is rejected. If the vote is valid, it’s added to the pool of unconfirmed transactions, waiting to be incorporated into a block.

The block mining step is initiated when there's a sufficient number of unconfirmed transactions or after a specified time interval. A set of transactions from the unconfirmed pool is selected to form a new block, referencing the previous block's hash to link the new block to the existing chain. The proof of work process begins by initializing a nonce to zero and computing the block's hash. If the hash does not meet the specified criteria (e.g., starting with '00'), the nonce is incremented, and the hash is recomputed. Once the criteria are met, the block is finalized by setting its nonce and hash, and then the new block is added to the blockchain. The unconfirmed transactions that were just confirmed by being added to the new block are then cleared.

Following this, a consensus mechanism ensures all nodes agree on the current state of the blockchain. Each peer node's version of the blockchain is reviewed, and if a node has a longer and valid blockchain, it replaces the current one. The newly block is announced to all peer nodes. Each node validates the new block, and if deemed valid, adds it to their version of the blockchain.

Finally, process ultimately ends up with the dissemination of the updated blockchain, where the new vote is embedded in it, exemplifying an electoral process open, secure, and hence democratic.



FIGURE 5. Shows the process “gesture-based virtual keyboard interface.

Figure 5 illustrates a user interface for our proposed electronic voting system (EVS) that utilizes a virtual, touchless interaction method. This virtual interaction could be part of an advanced user input method where users can engage with the system via hand gestures or other physical movements, such as eyebrow or nose gestures, without the need for physical contact. The specific highlighting of the letter ‘N’ on the virtual keyboard may be part of an authentication sequence or could represent an action within the voting process. The gesture paths leading to ‘N’ indicate the system’s ability to recognize complex gesture patterns, which can enhance user interaction by making it more intuitive and efficient,

particularly for users with physical limitations that prevent the use of traditional input devices.

V. RESULT AND DISCUSSION

We offer a comprehensive assessment of the proposed system performance via: subjective and objective measures. Through the subjective study, three questionnaires were analyzed with a view of ascertaining how the users viewed the system in which ethical approval was granted by the University of Diyala. These questionnaires are aimed at evaluating the ease of use, user satisfaction and speed. The participants were requested to give feedback on the general experience, user friendliness, and effectiveness in fulfilling their requirements. At the same time, the objective study focused on quantitative metrics of the system, using accuracy, precision, recall and F1-score measures. Integrating the subjective user perspectives with objective performance indices, we are able to provide a comprehensive understanding of the strengths and potential areas for improvement in our system.

It should be noted that the fundamental hardware and software components for implementing the system include:

1-Hardware:

- Camera: A high-definition webcam capable of shooting at 1280×720 -pixel resolution.
- Computer: With an Intel i7 and 16GB of RAM, supplemented with a dedicated NVIDIA GTX 1050 Ti to process the video feed in real-time.

2-Software:

- Programming Language: Python 3.8.
- Libraries: OpenCV for video capturing and processing, dlib for face detection and landmark prediction, cvzone for hand detection, and pynput for the virtual keyboard.
- Custom application: A Flask web application that integrates the libraries above in a way such that they provide a user interface and handle video streaming.

A. SUBJECTIVE STUDY

In this study, we use a comprehensive approach to evaluate the user experience and the perceptions about our system. The subjective study involves the administration of three unique questionnaires, each one is designed to capture important aspects of user interaction.

The first question in the subjective study aims to gauge users’ overall satisfaction with the proposed electronic voting system in which the users are asked.

How satisfied they are with the overall experience of using the proposed electronic voting system. Tables 3 shows the satisfaction levels by the users for the proposed electronic voting system, while Table 4 shows Percentage Distribution: Healthy vs. Disabled Individuals.

Our results showed that satisfaction is high across all groups, ranging from 61% to 75.5%, this suggests positive user experiences overall. Healthy users show slightly higher satisfaction (75.5% vs. 70% for total users, and 70.5% vs. 61% for seniors), while Senior Users, both healthy and disabled, exhibit good satisfaction results.

TABLE 3. User satisfaction levels with proposed electronic voting system.

Age Category	18-55						55+					
Response Age Category	45			20			17			18		
Category	Healthy			Disability			Healthy			Disability		
Answer Category	S	N	D	S	N	D	S	N	D	S	N	D
Response Answer	3	5	6	14	2	4	1	3	2	11	2	5
Age Response	65						35					
Total user	100											

TABLE 4. Percentage distribution: Healthy vs. Disabled individuals.

18-55				55+			
Healthy		Disability		Healthy		Disability	
S	75.5%	S	70%	S	70.5%	S	61%
N	11.11%	N	10%	N	17.6	N	11.11%
D	13.33%	D	20%	D	11.76%	D	27.7%

Dissatisfaction rates are generally low across all groups (ranging from 11.76% to 27.7%). This reinforces the positive sentiment towards the proposed electronic voting system.

In the second questionnaire, the crucial aspects of user comfort were investigated. Specifically, participants are prompted to articulate their comfort levels while engaging with the electronic voting system.

TABLE 5. Settings and parameters of the gesture recognition system.

Parameter/Setting	Description
Resolution	1280x720 pixels (Width x Height).
Face Detector	Dlib's frontal face detector for identifying faces in the video frame.
Landmark Predictor	Dlib's shape predictor using "shape_predictor_68_face_landmarks.dat" for detecting facial landmarks.
Hand Detector	CVzone's HandDetector with a detection confidence threshold set to 0.8.
Configuration	
Virtual Keyboard Layout	A 4-row layout including letters Q-P, A-semicol (;), Z-/ and special inputs ("Right eyebrow", "Left eyebrow", "Nose", "Fingers").
Button Size and Position	Each button is sized 85x85 pixels, positioned with an initial offset of 50 pixels from the frame's edge, and spaced by 100 pixels apart.
Interaction Mechanism	Uses Dlib for eyebrow and nose detection in facial recognition; CVzone HandDetector for hand and finger gesture recognition.
Gesture Detection Frequency	Facial gesture detection checks every 30 frames to prevent rapid unintended inputs.
Hand Gesture Detection Threshold	The distance between fingertips (index and thumb) less than 40 pixels is considered a "click" on the virtual keyboard.
Sleep Duration After Input	1 second sleep after detecting an input to prevent multiple unintended inputs.
Transparency Level for Buttons	Alpha blending value of 0.5 for drawing semi-transparent buttons over the video frame.

The table 5 summarizes the settings and parameters of the gesture recognition system using video processing and

biometric detection technologies. It includes hand gesture and face detection settings, face gesture detection settings, virtual keyboard mechanics settings, and visual feedback.

Table 6 shows the user comfort level when using the proposed system, while Table 7 shows Percentage Breakdown: Health and Disability Status by Age.

TABLE 6. User comfort level with proposed electronic voting system.

Age Category	18-55						55+					
Response Age Category	45		20				17			18		
Category	Healthy		Disability				Healthy			Disability		
Answer Category	C	N	U C	C	N	U C	C	N	U C	C	N	U C
Response Answer	35	5	5	18	1	1	12	2	3	1 4	1	3
Age Response	65						35					
Total user	100											

TABLE 7. Percentage breakdown: Health and disability status by age.

18-55				55+			
Healthy		Disability		Healthy		Disability	
C	77.77%	C	90%	C	70.5%	C	77.7%
N	11.11%	N	5%	N	11.76%	N	5.55%
UC	11.11%	UC	5%	UC	17.6%	UC	16.66%

Our results indicated that comfort levels are high across all groups, except for healthy senior users, this suggests a generally positive user experience in terms of usability and design.

Users with disabilities seem significantly more comfortable (90% vs. 77.77%). This could indicate better accessibility features, clearer design elements, or positive perceptions about the proposed electronic voting system.

Neutral responses remain low across all groups (ranging from 5% to 11.76%), this suggests most users have a clear comfort level, either positive or negative.

In general, levels of uncomfortable responses (ranging from 5% to 17.6%) In the third questionnaire, we specifically address the critical aspect of speed in the proposed electronic voting system. Participants are prompted to share their perspectives on how they rate the speed of the voting process.

Table 8 shows the user speed evaluation results while Table 9 shows Health and Disability Percentage Breakdown Across Age Segments.

The results we gained indicated that Overall, performance seems acceptable to good across all groups, with "fast" response rates ranging from 77.7% to 85%. This suggests users are experiencing generally swift interaction, Users with disabilities appear to encounter slightly fewer "slow" responses compared to healthy users (5% vs. 11.11%). Neutral responses remain generally low across all groups (ranging from 5.8% to 11.11%).

TABLE 8. Percentage breakdown: Health and disability status by age.

Age Category	18-55						55+					
Response Age Category	45			20			17			18		
Category	Healthy			Disability			Healthy			Disability		
Answer Category	F	N	S	F	N	S	F	N	S	F	N	S
Response Answer	36	4	5	17	2	1	14	1	2	14	2	2
Age Response	65						35					
Total user	100											

TABLE 9. Health and disability percentage breakdown across age segments.

18-55			55+		
Healthy		Disability	Healthy		Disability
F	80%	F 85%	F	82.3%	F 77.7%
N	8.8%	N 10%	N	5.8%	N 11.11%
S	11.11%	S 5%	S	11.76%	S 11.11%

This suggests most users have a clear opinion on performance, either positive or negative.

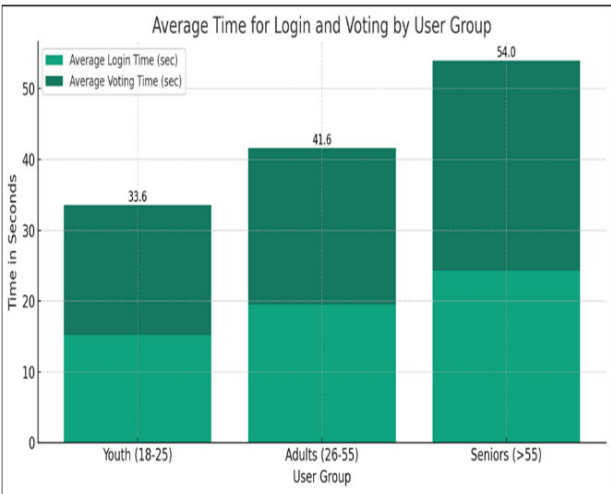


FIGURE 6. The user interaction times.

B. USER INTERACTION TIMES

Figure 6 visualizes the average time it takes different user groups to log in and vote using the proposed system. It clearly shows that as age increases, the interaction time will also increase, with seniors taking the longest. This aligns with expectations that older users might have slower interaction speeds, potentially due to less familiarity with technology or slower motor skills. Additionally, network complexity can impact real-time performance, and factors like server response time and data processing can contribute to these results. Comparing login and voting times, it is observed that the average time to vote is longer than the average time to

log in for all user groups. This implies that the voting process requires more time than the login process. The most significant difference in average time between voting and login is noted among seniors, with a margin of 5.4 seconds. This indicates that the voting process may pose greater difficulty for seniors compared to younger users.

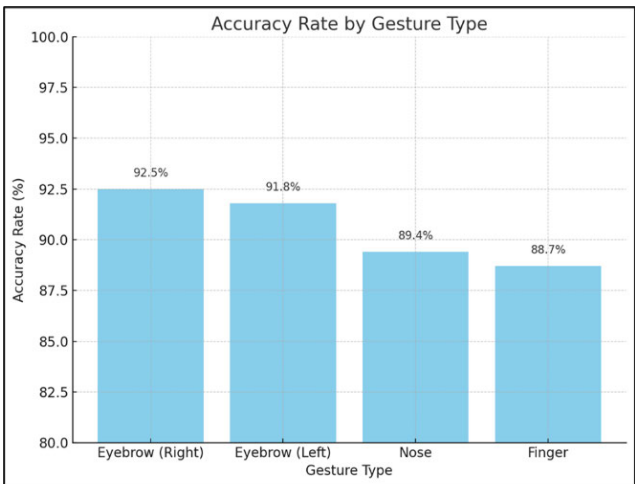


FIGURE 7. The accuracy rate by gesture type.

C. ACCURACY RATE BY GESTURE TYPE

Figure 7. illustrates the accuracy rates for different types of gesture recognition in the proposed system. Right eyebrow gestures have the highest accuracy at 92.5%, closely followed by left eyebrow gestures at 91.8%. Nose gestures exhibit a slightly lower accuracy rate of 89.4%, while finger gestures have the lowest accuracy at 88.7%. The study has shown that facial expressions, especially eyebrow movement, are recognized with more accuracy compared to finger movement, and perhaps this difference of accuracy was because facial movement is more easy and distinct to be captured, unlike the complexities involved in finger movement movements.

D. CONFUSION MATRIX RESULT

Figure 8. indicates That our proposed system’s confusion matrix shows its excellent performance in discriminating between real and fake voters, with 99 actual voters correctly accepted as true positives and 30 fraudulent voters effectively rejected as true negatives. This illustrates the system’s significant ability to provide a secure and trustworthy voting environment by accurately recognizing legitimate participants. However, a single false negative, in which an actual voter was incorrectly excluded, indicates a minor limitation, emphasizing the system’s challenge. However, the system’s precision is bolstered by the absence of false positives, proving its severe steps to ensure that no fake votes are incorrectly accepted. this delicate balance between high accuracy in voter verification and a tiny margin of error, emphasizes the proposed system’s significance in ensuring the integrity and dependability of the voting process, making it an essential tool in the democratic framework.

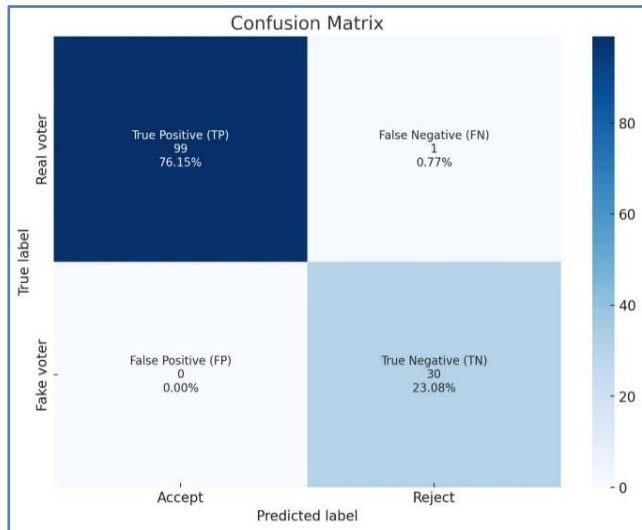


FIGURE 8. Shows the confusion matrix results for the proposed system.

TABLE 10. Accuracy and error rates.

User Group	Accuracy (%)	EER (%)	FAR (%)	FRR (%)
Youth	100	N/A	0	0
Adults	100	N/A	0	0
Seniors	99.23	~2.86	0	2.86

Table 10. presents accuracy and error rates for three age groups using a voting system. Both Youth and Adults have perfect accuracy with no false rejections or acceptances, indicated by 100% accuracy and 0% Equal Error Rate (EER), False acceptance Rate (FAR), and False rejection Rate (FRR). For Seniors, the accuracy slightly drops to 99.23% due to one instance of false rejection, resulting in an estimated EER and FRR of approximately 2.86%, while no false acceptances were recorded, maintaining a 0% FAR. These metrics suggest the system is highly accurate.

Table 11 compares our proposed electronic voting system to existing systems. It emphasizes technology integration, with the proposed solution combining gesture recognition and blockchain, as opposed to related works that focus exclusively on blockchain or authentication methods separately.

The proposed system also prioritizes accessibility and inclusivity, targeting a broad spectrum of voters, including those with disabilities, which are frequently disregarded in other studies. In terms of user interface and experience, the proposed system provides a user-friendly and adaptive interface, addressing critiques of complex interfaces in other works that may hinder participation.

Finally, the proposed system includes novel authentication methods that use gesture-based inputs such as brow or nose motions, giving a more inclusive and adaptive approach than previous research, which relied on standard biometrics or multi-factor authentication.

TABLE 11. Comparison proposed electronic voting system with the state-of-the-art.

Aspect	Related work	Proposed system
Integration of Technologies	Studies focus mainly on blockchain ([21], [22],[24]) or advanced authentication methods ([26], [27],[28]) separately.	Combines gesture recognition with blockchain for enhanced security and user interaction.
Accessibility and Inclusivity	Often overlooks the needs of diverse voters, including those with disabilities ([21], [22], [23],[26]).	Prioritizes inclusivity with gesture-based interfaces for a wider range of users, including those with disabilities.
User Interface and Experience	Critiqued for complex interfaces ([25], [26], [28]) potentially deterring participation.	Features a user-friendly and adaptable interface, catering to all users, including the visually impaired and technology-averse.
Security and Transparency	Focuses on voting integrity using blockchain ([22], [23], [25]) but not always with user-friendly authentication.	Enhances security with blockchain and simplifies authentication with innovative gesture-based methods.
Innovation in Authentication	Relies on traditional biometrics or multi-factor authentication ([27], [28], [29],[30]) which can be challenging for some users.	Introduces gesture-based authentication (eyebrow or nose movements) for a more inclusive and adaptable approach.

VI. CONCLUSION AND FUTURE WORKS

The integration of gesture recognition and blockchain technology in the proposed electronic voting system presents a remarkable advancement, as evidenced by its evaluation results. This system provides a greater security and transparency due to the immutable nature of blockchain technology, as well as high levels of accuracy and reliability in the gesture recognition. Scores of satisfactions on the proposed electronic voting system are high (61%-75.5%) as well as the positive feedbacks regarding comfortability and speed. The system has a high-performance level, low user dissatisfaction rates and thus has a positive implementation prospect with possible target improvements in the design of accessibility features for certain groups of users. Furthermore, the evaluation results indicate a high accuracy rate for several gestures which makes it easy to use the system for a wide range of electorates that include the people with disabilities and the elderly. In addition, the user-friendly interface of the system on one hand and its inclusive design on the other are confirmed by security metrics-high precision, recall and also

extremely low Equal Error Rate (EER). These findings collectively demonstrate the potential of this system to transform the voting procedure, making it secure, accessible and user centered, thus addressing the key issues of the conventional methods as well as the current electronic voting mechanisms. The future work include, improvements may include the integration of emotion recognition to allow easier user interaction, especially for the people who cannot use conventional input methods. Investigation of other more sophisticated biometric verification methods like the vein pattern recognition may provide an easy and secure authentication for a wide number of users.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

The background research, conceptual framework, methodology design, data collection, system implementation, result analysis and comparison, draft preparation, and visualization were all conducted by the first author. Meanwhile, the second and third authors were responsible for supervising the project, reviewing the work, and managing the overall administration of the project.

REFERENCES

- [1] W. Wahid and T. S. Noushin, "Enhancing trust and integrity in electronic voting: A blockchain-based approach," in *Proc. Conf., CyberShield, Defending Digit. Landscape*, Islamabad, Pakistan, 2023.
- [2] S. Park, M. Specter, N. Narula, and R. L. Rivest, "Going from bad to worse: From internet voting to blockchain voting," *J. Cybersecur.*, vol. 7, no. 1, Feb. 2021, Art. no. tyaa025, doi: [10.1093/cybsec/tyaa025](https://doi.org/10.1093/cybsec/tyaa025).
- [3] F. Hjalmarsson, G. K. Hreioarsson, M. Hamdaqa, and G. Hjalmtýsson, "Blockchain-based e-voting system," in *Proc. IEEE 11th Int. Conf. Cloud Comput. (CLOUD)*, Jul. 2018, pp. 983–986, doi: [10.1109/CLOUD.2018.00151](https://doi.org/10.1109/CLOUD.2018.00151).
- [4] N. Kshetri and J. Voas, "Blockchain-enabled e-voting," *IEEE Softw.*, vol. 35, no. 4, pp. 95–99, Jul. 2018, doi: [10.1109/MS.2018.2801546](https://doi.org/10.1109/MS.2018.2801546).
- [5] D. T. Kadhim and B. S. Mahdi, "Enhancing the security of the blockchain and the file contents," in *Proc. 3rd Inf. Technol. Enhance e-Learning Appl. (IT-ELA)*, Dec. 2022, pp. 6–11.
- [6] D. Willis, T. Merivaki, and I. Ziogas, "Election data transparency: Obtaining precinct-level election returns," *Public Integrity*, vol. 24, no. 2, pp. 162–178, Mar. 2022, doi: [10.1080/10999922.2021.1883854](https://doi.org/10.1080/10999922.2021.1883854).
- [7] Y. Liu and Q. Wang, "An e-voting protocol based on blockchain," *IACR Cryptol. Cryptol. ePrint Arch.*, pp. 1043–1045, Oct. 2017.
- [8] P. Raesko, "Blockchain and democracy," *Soc. Economy*, vol. 41, no. 3, pp. 353–369, Sep. 2019, doi: [10.1556/204.2019.007](https://doi.org/10.1556/204.2019.007).
- [9] U. Jafar, M. J. A. Aziz, and Z. Shukur, "Blockchain for electronic voting system—Review and open research challenges," *Sensors*, vol. 21, no. 17, p. 5874, Aug. 2021, doi: [10.3390/s21175874](https://doi.org/10.3390/s21175874).
- [10] B. S. Mahdi, M. J. Hadi, and A. R. Abbas, "Intelligent security model for password generation and estimation using hand gesture features," *Big Data Cognit. Comput.*, vol. 6, no. 4, p. 116, Oct. 2022, doi: [10.3390/bdcc6040116](https://doi.org/10.3390/bdcc6040116).
- [11] O. Y. Fadhil, B. S. Mahdi, and A. R. Abbas, "Using VGG models with intermediate layer feature maps for static hand gesture recognition," *Baghdad Sci. J.*, vol. 20, no. 5, p. 1808, 2023.
- [12] K. Garg, P. Saraswat, S. Bisht, S. Kr. Aggarwal, S. K. Kothuri, and S. Gupta, "A comparative analysis on e-voting system using blockchain," in *Proc. 4th Int. Conf. Internet Things, Smart Innov. Usages (IoT-SIU)*, Apr. 2019, pp. 1–4, doi: [10.1109/IoT-SIU.2019.8777471](https://doi.org/10.1109/IoT-SIU.2019.8777471).
- [13] E. Febriyanto, N. Rahayu, K. Pangaribuan, and P. A. Sunarya, "Using blockchain data security management for e-voting systems," in *Proc. 8th Int. Conf. Cyber IT Service Manage. (CITSM)*, Oct. 2020, pp. 1–4, doi: [10.1109/CITSM50537.2020.9268847](https://doi.org/10.1109/CITSM50537.2020.9268847).
- [14] C. K. Adiputra, R. Hjort, and H. Sato, "A proposal of blockchain-based electronic voting system," in *Proc. 2nd World Conf. Smart Trends Syst., Secur. Sustainability (WorldS4)*, Oct. 2018, pp. 22–27, doi: [10.1109/WorldS4.2018.8611593](https://doi.org/10.1109/WorldS4.2018.8611593).
- [15] S. Bláhová, "Online democracy: Applying Hannah Arendt's model of democracy to the internet," *Theoria*, vol. 89, no. 6, pp. 856–871, Dec. 2023, doi: [10.1111/theo.12501](https://doi.org/10.1111/theo.12501).
- [16] B. Wang, J. Sun, Y. He, D. Pang, and N. Lu, "Large-scale election based on blockchain," *Proc. Comput. Sci.*, vol. 129, pp. 234–237, Jan. 2018, doi: [10.1016/j.procs.2018.03.063](https://doi.org/10.1016/j.procs.2018.03.063).
- [17] M. Pawlak and A. Poniszewska-Maranda, "Trends in blockchain-based electronic voting systems," *Inf. Process. Manage.*, vol. 58, no. 4, Jul. 2021, Art. no. 102595, doi: [10.1016/j.ipm.2021.102595](https://doi.org/10.1016/j.ipm.2021.102595).
- [18] C. Killer, B. Rodrigues, E. J. Scheid, M. Franco, M. Eck, N. Zaugg, A. Scheitlin, and B. Stiller, "Provotum: A blockchain-based and end-to-end verifiable remote electronic voting system," in *Proc. IEEE 45th Conf. Local Comput. Netw. (LCN)*, Nov. 2020, pp. 172–183, doi: [10.1109/LCN48667.2020.9314815](https://doi.org/10.1109/LCN48667.2020.9314815).
- [19] D. Kirillov, V. Korkhov, V. Petrunin, M. Makarov, I. M. Khamitov, and V. Dostov, "Implementation of an e-voting scheme using hyperledger fabric permissioned blockchain," in *Computational Science and Its Applications—ICCSA*, Cham, Switzerland: Springer, 2019, pp. 509–521, doi: [10.1007/978-3-030-24296-1](https://doi.org/10.1007/978-3-030-24296-1).
- [20] A. Leema, A. Zameer, and P. Gulzar, "Trusted and secured e-voting election system based on block chain technology," in *Proc. Int. Conf. Comput. Netw., Big Data IoT (ICCB)*, Cham, Switzerland: Springer, 2020, pp. 81–88, doi: [10.1007/978-3-030-43192-1_9](https://doi.org/10.1007/978-3-030-43192-1_9).
- [21] B. M. B. Pereira, J. M. Torres, P. M. Sobral, R. S. Moreira, C. P. D. A. Soares, and I. Pereira, "Blockchain-based electronic voting: A secure and transparent solution," *Cryptography*, vol. 7, no. 2, p. 27, May 2023, doi: [10.3390/cryptography7020027](https://doi.org/10.3390/cryptography7020027).
- [22] B. A. K. Vinsura, R. M. V. D. Bandara, A. D. A. I. Gunasekara, B. Hettige, and G. A. I. Uwanthika, "UxVote-blockchain-based e-voting system for secure electronic voting," in *Proc. 15th Int. Res. Conf.—Gen. Sir John Kotelawala*, Sri Lanka, 2022.
- [23] M. S. Farooq, U. Iftikhar, and A. Khelifi, "A framework to make voting system transparent using blockchain technology," *IEEE Access*, vol. 10, pp. 59959–59969, 2022, doi: [10.1109/ACCESS.2022.3180168](https://doi.org/10.1109/ACCESS.2022.3180168).
- [24] C. Anwar ul Hassan, M. Hammad, J. Iqbal, S. Hussain, S. S. Ullah, H. AlSalman, M. A. A. Mosleh, and M. Arif, "A liquid democracy enabled blockchain-based electronic voting system," *Sci. Program.*, vol. 2022, pp. 1–10, Jan. 2022, doi: [10.1155/2022/1383007](https://doi.org/10.1155/2022/1383007).
- [25] S. T. Alvi, M. N. Uddin, L. Islam, and S. Ahamed, "DVTChain: A blockchain-based decentralized mechanism to ensure the security of digital voting system voting system," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 34, no. 9, pp. 6855–6871, Oct. 2022, doi: [10.1016/j.jksuci.2022.06.014](https://doi.org/10.1016/j.jksuci.2022.06.014).
- [26] X. Yang, Y. Zhang, S. Wang, B. Yu, F. Li, Y. Li, and W. Yan, "LedgerDB: A centralized ledger database for universal audit and verification," *Proc. VLDB Endowment*, vol. 13, no. 12, pp. 3138–3151, Aug. 2020, doi: [10.14778/3415478.3415540](https://doi.org/10.14778/3415478.3415540).
- [27] X. Yang, R. Zhang, C. Yue, Y. Liu, B. C. Ooi, Q. Gao, Y. Zhang, and H. Yang, "VeDB: A software and hardware enabled trusted relational database," *Proc. ACM Manage. Data*, vol. 1, no. 2, pp. 1–27, Jun. 2023, doi: [10.1145/3589774](https://doi.org/10.1145/3589774).
- [28] B. A. Oke, O. M. Olaniyi, A. A. Aboaba, and O. T. Arulogun, "Multifactor authentication technique for a secure electronic voting system," *Electron. Government, Int. J.*, vol. 17, no. 3, p. 312, 2021, doi: [10.1504/eg.2021.115999](https://doi.org/10.1504/eg.2021.115999).
- [29] E. H. Piam, A. Mahmud, R. Abdulhafor, S. Wani, A. Ibrahim, and A. Olowolayemo, "Face authentication-based online voting system," *Int. J. Perceptive Cogn. Comput.*, vol. 8, no. 1, pp. 19–23, 2022.
- [30] M. Ahmad, A. U. Rehman, N. Ayub, M. Alshehri, M. A. Khan, A. Hameed, and H. Yetgin, "Security, usability, and biometric authentication scheme for electronic voting using multiple keys," *Int. J. Distrib. Sensor Netw.*, vol. 16, no. 7, Jul. 2020, Art. no. 155014772094402, doi: [10.1177/1550147720944025](https://doi.org/10.1177/1550147720944025).
- [31] O. S. Adewale, K. Olutayo, and S. E. Boyinbode, "An innovative approach in electronic voting system based on fingerprint and visual semagram," *Int. J. Inf. Eng. Electron. Bus.*, vol. 13, no. 5, pp. 24–37, Oct. 2021, doi: [10.5815/ijeeb.2021.05.03](https://doi.org/10.5815/ijeeb.2021.05.03).
- [32] S. J. J. Arputhamoni and A. G. Saravanan, "Online smart voting system using biometrics based facial and fingerprint detection on image processing and CNN," in *Proc. 3rd Int. Conf. Intell. Commun. Technol. Virtual Mobile Netw. (ICICV)*, Feb. 2021, pp. 1–7, doi: [10.1109/ICICV50876.2021.9388405](https://doi.org/10.1109/ICICV50876.2021.9388405).

- [33] M. T. Abdulhadi and A. R. Abbas, "Human action behavior recognition in still images with proposed frames selection using transfer learning," *Int. J. Online Biomed. Eng. (iJOE)*, vol. 19, no. 6, pp. 47–65, May 2023.
- [34] M. Xu, D. Chen, and G. Zhou, "Real-time face recognition based on dlib," in *Innovative Computing (Lecture Notes in Electrical Engineering)*. Singapore: Springer, 2020, pp. 1451–1459, doi: [10.1007/978-981-15-5959-4_177](https://doi.org/10.1007/978-981-15-5959-4_177).
- [35] I. Putra, "Penerapan library tensorflow, cvzone, dan numpy pada sistem deteksi bahasa isyarat secara real time," *Jurnal Krisnadana*, vol. 2, no. 3, pp. 412–423, 2023, doi: [10.58982/krisnadana.v2i3.335](https://doi.org/10.58982/krisnadana.v2i3.335).



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